

Product Vision Document

Project: Aarvii CCTV AMC Management System **Phase:** D0 — Project Foundation Documentation **Source of truth:** requirements-freeze-v1.md (Approved & Frozen, V1) **Status:** Derived documentation — no requirement changes introduced.

1. Business vision

Build a **centralized CCTV AMC (Annual Maintenance Contract) Management System** for **Aarvii Technologies** that manages the complete AMC lifecycle in one platform:

- Public website (www.aarvii.in) with lead generation
- Customers, sites, and asset summaries
- AMC plans and AMC contracts (with renewal terms)
- Preventive maintenance scheduling and service visits
- Complaint tickets
- Engineers and their field activities
- Invoices
- Customer self-service portal and Engineer portal (web + mobile)

The platform provides **complete visibility** of the customer AMC lifecycle, engineer activities, service history, contract renewals, invoices, and complaint resolution (freeze \$1).

The system is delivered as a **business module suite** on top of the frozen Ashraak Enterprise Platform V1, reusing Authentication, Roles, Permissions, Files, Notifications, Audit, Theme Engine, Mobile Foundation, API Keys, and Webhooks with **no duplicate implementation** (freeze \$20).

2. Objectives

#	Objective	Freeze reference
O1	Convert public website traffic into qualified leads and AMC customers	\$2 (Public Website), \$10 (Lead Lifecycle)
O2	Maintain a single source of truth for customers, sites, and asset summaries	\$5, \$6, \$7
O3	Manage AMC contracts with a permanent master record and renewal-term history	\$8
O4	Automate preventive maintenance visit generation from AMC plan frequency	\$11
O5	Enforce verifiable field service evidence (selfie, GPS, photos, signature)	\$12
O6	Gate customer-visible reports behind admin approval	\$13
O7	Provide end-to-end complaint ticket management with reopen capability	\$14
O8	Issue and track invoices per AMC contract term with PDF download	\$16, \$19
O9	Keep customers and engineers informed via Email and SMS notifications	\$17
O10	Deliver self-service via Customer and Engineer portals on web and Flutter mobile apps	\$2, \$18

3. Problem statement

Without a centralized system, a CCTV AMC business faces:

- **Lost leads** — website inquiries and quote requests are not captured or tracked through a sales pipeline.
- **Fragmented customer data** — customers, sites, contacts, and installed asset counts live in spreadsheets or on paper.
- **Untracked AMC lifecycle** — contract renewals and plan history are not preserved; expiries are missed.
- **Unmanaged preventive maintenance** — visits owed under AMC frequency are not planned, assigned, or verified.
- **No field accountability** — no proof an engineer attended a site (selfie, GPS, photos, customer signature).
- **Opaque complaint handling** — tickets lack lifecycle, priority, assignment, and reopen tracking.
- **Manual invoicing** — invoices are not tied to contract terms; customers cannot self-serve downloads.
- **No customer self-service** — customers must call to learn AMC status, visit dates, or invoice details.

The Aarvii CCTV AMC Management System addresses all of the above within the approved V1 scope.

4. Success criteria

V1 is successful when:

#	Criterion
S1	Every website inquiry / quote request automatically creates a lead (freeze §10)
S2	A lead can be converted in one flow into Customer + Site + Initial AMC Contract (freeze §10)
S3	Each site holds at most one active AMC contract; renewal history is preserved as terms (freeze §6, §8)
S4	Scheduled visits are auto-generated from AMC plan frequency, with mandatory engineer assignment (freeze §11)
S5	No visit can be completed without selfie, GPS, ≥1 photo, customer signature, and remarks (freeze §12)
S6	Customers see visit reports only after admin approval (freeze §13)
S7	Tickets follow the approved lifecycle and can be raised by Customer, Admin, or Engineer, and reopened by Customer (freeze §14)
S8	Invoices are linked to AMC contract terms and downloadable as PDF by customers (freeze §16)
S9	All approved notification events fire on Email and SMS (freeze §17)
S10	Customer and Engineer apps (Flutter) ship with the approved feature sets, including engineer offline support (freeze §18)
S11	AMC Contract, Visit Report, and Invoice PDFs are generated (freeze §19)
S12	Zero duplicate implementation of base platform capabilities (freeze §20)

5. Target users

The four approved actors (freeze §3); detailed personas in [actors-and-personas.md](#):

Actor	Description
Public Visitor	Anonymous website user browsing services, submitting inquiries and quote requests
Customer	Contract holder using the Customer Portal / App for AMC, tickets, invoices, and profile

Actor	Description
Engineer	Field service engineer executing visits and tickets via the Engineer Portal / App
Admin	Aarvii back-office staff managing all modules through the Admin Portal

6. Expected benefits

Beneficiary	Benefits
Aarvii Technologies (business)	Captured leads and conversion pipeline; preserved AMC renewal history; automated visit planning; verified field work; ticket accountability; invoice traceability; reporting across the operation
Customers	Self-service visibility of AMC, visits, service history, tickets, and invoices; ability to raise/reopen tickets and request renewals; PDF invoice downloads
Engineers	Clear assigned work queue; structured mobile reporting (photos, GPS, selfie, signature); offline support in the field
Admins	Single console for leads, customers, sites, assets, plans, contracts, scheduling, tickets, engineers, invoices, and reports; approval control over customer-visible reports

7. Guardrails

- The scope is **frozen** per [requirements-freeze-v1.md §22](#); changes only via approved change request.
- Out-of-scope items (freeze §21) are explicitly excluded from this vision — see [scope-freeze-v1.md](#).
- All design and build phases (BRD, HLD, ERD, LLD, APIs, UI, Mobile, Development) must trace back to the freeze document.

Related documents

- [scope-freeze-v1.md](#)
- [business-requirements-document.md](#)
- [high-level-design.md](#)
- [project-roadmap.md](#)
- Platform discovery: [../project-bootstrap/platform-discovery-report.md](#)